Name: G.Varshith Reddy

Enrollment_number:2303A52287

Batch:37

PROMPT ENGINEERING TASK REPORT

Task 1: Cleaning Sensor Data

Prompt:

Give a function that filters out all the negative numbers from a list.**Explanation** As the prompt becomes more specific, the calculator logic improves by handling operators correctly and preventing division by zero.

Code

```
def filter_negative_numbers(numbers):
    """
    Filters out all negative numbers from a list.

    Args:
        numbers (list): List of numbers to filter

    Returns:
        list: List containing only non-negative number
    """
    return [num for num in numbers if num >= 0]

# Example usage:
# numbers = [-5, 3, -1, 0, 8, -2, 10]
# filtered = filter_negative_numbers(numbers)
# print(filtered)  # Output: [3, 0, 8, 10]
```

Output

```
Original array: [-5, 3, -1, 0, 8, -2, 10]
Filtered array: [3, 0, 8, 10]
```

Task 2: String Character Analysis

Prompt

giva a function that counts vowels, consonants,and digits in a string

Code

```python
def count_string_components(text):
    vowels = "aeiouAEIOU"
    vowel_count = 0
    consonant_count = 0
    digit_count = 0

    for char in text:
        if char.isdigit():
            digit_count += 1
        elif char.isalpha():
            if char in vowels:
                vowel_count += 1
            else:
                consonant_count += 1

    return vowel_count, consonant_count, digit_count
```

Output

```
Original array: [-5, 3, -1, 0, 8, -2, 10]
Filtered array: [3, 0, 8, 10]
String: 'Hello World 123!'
Vowels: 3
Consonants: 7
Digits: 3
```

Task 3: Palindrome Check – Tool Comparison

Prompt

give a function that checks the palindrome

.

Code

```python
def is_palindrome(text):
    """
    # Convert to lowercase and keep only alphanumeric characters
    cleaned = ''.join(char.lower() for char in text if char.isalnum())

    # Check if cleaned string equals its reverse
    return cleaned == cleaned[::-1]
```

Output

```
'racecar' -> True
'A man a plan a canal Panama' -> True
'hello' -> False
'Madam' -> True
'12321' -> True
'12345' -> False
```

Task 4: Code Explanation Using AI

Prompt

Explain the palindrome code

Code

```
Check if a given text is a palindrome.
This function determines whether the input text reads the same forwards and backwards,
ignoring case differences, spaces, and non-alphanumeric characters.
Args:
    text (str): The input string to check for palindrome property.
Returns:
    bool: True if the text is a palindrome, False otherwise.
Example:
    >>> is_palindrome("A man a plan a canal Panama")
    True
    >>> is_palindrome("race a car")
    False
    >>> is_palindrome("Was it a car or a cat I saw?")
    True
Note:
    The function performs case-insensitive comparison and ignores all
    non-alphanumeric characters (spaces, punctuation, etc.) when
    determining if the text is a palindrome.
```